

5. Równania kwadratowe 3

Wyznacz miejsca zerowe funkcji f, g, h:

a) $f(x) = 2x^2 - 8x + 5$; $g(x) = 2(x - 0,25)^2 - 8(x - 0,25) + 5$;

$h(x) = 2\left(x - \frac{\sqrt{6}}{2}\right)^2 - 8\left(x - \frac{\sqrt{6}}{2}\right) + 5$

b) $f(x) = -0,5x^2 + 4x + 6$; $g(x) = -0,5(x + 3,5)^2 + 4(x + 3,5) + 6$;

$h(x) = -0,5(x + 2\sqrt{7})^2 + 4(x + 2\sqrt{7}) + 6$

Kiedy we wzorze powstana się to samo wyrażenie więcej niż raz, można za to wyrażenie podstawić tzw. zmienną pomocniczą, co pozwala uprościć równanie.

a) $f(x) = 2x^2 - 8x + 5$
 $g(x) = 2(x - 0,25)^2 - 8(x - 0,25) + 5$
 $h(x) = 2\left(x - \frac{\sqrt{6}}{2}\right)^2 - 8\left(x - \frac{\sqrt{6}}{2}\right) + 5$

① $f(x) = 2x^2 - 8x + 5$

$2x^2 - 8x + 5 = 0$ $\Delta = b^2 - 4 \cdot a \cdot c$

$a = 2$ $\Delta = (-8)^2 - 4 \cdot 2 \cdot 5 = 64 - 40 = 24$
 $b = -8$ $\sqrt{\Delta} = \sqrt{24} = 2\sqrt{6}$
 $c = 5$

$x_1 = \frac{-b - \sqrt{\Delta}}{2a} = \frac{-(-8) - 2\sqrt{6}}{2 \cdot 2} = \frac{8 - 2\sqrt{6}}{4} = 2 - \frac{\sqrt{6}}{2}$

$x_2 = \frac{-b + \sqrt{\Delta}}{2a} = \frac{-(-8) + 2\sqrt{6}}{2 \cdot 2} = \frac{8 + 2\sqrt{6}}{4} = 2 + \frac{\sqrt{6}}{2}$

② $g(x) = 2(x - 0,25)^2 - 8(x - 0,25) + 5$

niech $t = x - 0,25$

$2t^2 - 8t + 5 = 0$ $\Delta = b^2 - 4 \cdot a \cdot c$

$a = 2$ $\Delta = (-8)^2 - 4 \cdot 2 \cdot 5 = 64 - 40 = 24$
 $b = -8$ $\sqrt{\Delta} = \sqrt{24} = 2\sqrt{6}$
 $c = 5$

$t_1 = \frac{-b - \sqrt{\Delta}}{2a} = \frac{-(-8) - 2\sqrt{6}}{2 \cdot 2} = \frac{8 - 2\sqrt{6}}{4} = 2 - \frac{\sqrt{6}}{2}$

$t_2 = \frac{-b + \sqrt{\Delta}}{2a} = \frac{-(-8) + 2\sqrt{6}}{2 \cdot 2} = \frac{8 + 2\sqrt{6}}{4} = 2 + \frac{\sqrt{6}}{2}$

$t_1 = x_1 - 0,25 \Rightarrow x_1 = t_1 + 0,25$
 $x_1 = 2,25 - \frac{\sqrt{6}}{2}$

$t_2 = x_2 - 0,25 \Rightarrow x_2 = t_2 + 0,25$
 $x_2 = 2,25 + \frac{\sqrt{6}}{2}$

③ $h(x) = 2\left(x - \frac{\sqrt{6}}{2}\right)^2 - 8\left(x - \frac{\sqrt{6}}{2}\right) + 5$

niech $t = x - \frac{\sqrt{6}}{2}$

$2t^2 - 8t + 5 = 0 \Rightarrow t_1 = 2 - \frac{\sqrt{6}}{2}$
 $t_2 = 2 + \frac{\sqrt{6}}{2}$

$$t_1 = x_1 - \frac{\sqrt{6}}{2} \Rightarrow x_1 = t_1 + \frac{\sqrt{6}}{2}$$

$$x_1 = 2 - \frac{\sqrt{6}}{2} + \frac{\sqrt{6}}{2} = 2$$

$$t_2 = x_2 - \frac{\sqrt{6}}{2} \Rightarrow x_2 = t_2 + \frac{\sqrt{6}}{2}$$

$$x_2 = 2 + \frac{\sqrt{6}}{2} + \frac{\sqrt{6}}{2} = 2 + \sqrt{6}$$

odp: $f(x)=0$ dla $x=2-\frac{\sqrt{6}}{2}$; $x=2+\frac{\sqrt{6}}{2}$,
 $g(x)=0$ dla $x=2,25-\frac{\sqrt{6}}{2}$; $x=2,25+\frac{\sqrt{6}}{2}$,
 $h(x)=0$ dla $x=2$; $x=2+\sqrt{6}$

b) $f(x) = -0,5x^2 + 4x + 6$

$$g(x) = -0,5(x+3,5)^2 + 4(x+3,5) + 6$$

$$h(x) = -0,5(x+2\sqrt{7})^2 + 4(x+2\sqrt{7}) + 6$$

① $f(x) = -0,5x^2 + 4x + 6$

$$-\frac{1}{2}x^2 + 4x + 6 = 0 \quad | \cdot 2$$

$$-x^2 + 8x + 12 = 0$$

$$\Delta = b^2 - 4ac$$

$$a = -1 \quad \Delta = 8^2 - 4 \cdot (-1) \cdot 12 = 64 + 48 = 112$$

$$b = 8 \quad \sqrt{\Delta} = \sqrt{112} = 4\sqrt{7}$$

$$c = 12$$

$$x_1 = \frac{-b - \sqrt{\Delta}}{2a} = \frac{-8 - 4\sqrt{7}}{2 \cdot (-1)} = 4 + 2\sqrt{7}$$

$$x_2 = \frac{-b + \sqrt{\Delta}}{2a} = \frac{-8 + 4\sqrt{7}}{2 \cdot (-1)} = 4 - 2\sqrt{7}$$

② $g(x) = -0,5(x+3,5)^2 + 4(x+3,5) + 6$

Niech $t = x + 3,5$

$$-\frac{1}{2}t^2 + 4t + 6 = 0 \quad | \cdot 2$$

$$-t^2 + 8t + 12 = 0$$

$$t_1 = 4 + 2\sqrt{7} \quad t_2 = 4 - 2\sqrt{7}$$

$$t_1 = x_1 + 3,5 \Rightarrow x_1 = t_1 - 3,5$$

$$x_1 = 0,5 + 2\sqrt{7}$$

$$t_2 = x_2 + 3,5 \Rightarrow x_2 = t_2 - 3,5$$

$$x_2 = 0,5 - 2\sqrt{7}$$

③ $h(x) = -0,5(x+2\sqrt{7})^2 + 4(x+2\sqrt{7}) + 6$

Niech $t = x + 2\sqrt{7}$

$$-\frac{1}{2}t^2 + 4t + 6 = 0 \quad | \cdot 2$$

$$-t^2 + 8t + 12 = 0$$

$$t_1 = 4 + 2\sqrt{7} \quad t_2 = 4 - 2\sqrt{7}$$

$$t_1 = x_1 + 2\sqrt{7} \Rightarrow x_1 = t_1 - 2\sqrt{7}$$

$$x_1 = 4$$

$$t_2 = x_2 + 2\sqrt{7} \Rightarrow x_2 = t_2 - 2\sqrt{7}$$

$$x_2 = 4 - 4\sqrt{7}$$

Odp: $f(x)=0$ dla $x=4-2\sqrt{7}$, $x=4+2\sqrt{7}$,
 $g(x)=0$ dla $x=0,5-2\sqrt{7}$, $x=0,5+2\sqrt{7}$,
 $h(x)=0$ dla $x=4$, $x=4-4\sqrt{7}$